

Domain-Specific Text Analysis and Retrieval System

Consumer Mental-Health Guidance Corpus

Course: Natural Language Processing (DATA403), Vidyashilp University **Assignment:** AS1: Domain-Specific Text Analysis and Retrieval System **Team:** Tanmayee K M · Ahmed · Jeromi **Deliverables:** Domain_Text_Analysis_Retrieval.ipynb (all modules), results/ (CSV checkpoints, Tables A–J, inverted_index.json), app.py + src/nlp_pipeline.py (Gradio GUI), src/build_corpus.py (corpus pipeline)

Abstract

We built a complete text-analysis and retrieval system for **public mental-health guidance**: 17 documents from NICE, WHO, NIMH and NHS (41,297 words), covering depression, anxiety, panic disorder, PTSD, self-harm and suicide, medication, sleep, loneliness, stress and substance use. Every NLP step was first compared experimentally on this corpus: tokenizers, sentence splitters, BPE, stop lists, stemmers, lemmatizers, POS taggers and NER. The step was then **selected and ordered** for the domain. The resulting **Final Pipeline** is: number policy → custom regex tokenizer → lowercase → punctuation removal → spaCy lemmatization in context (with domain rules) → custom stop-word removal that keeps negation and modal verbs. It reduces the vocabulary by 46% while keeping safety-critical meaning ("do **not** stop taking" stays *not stop take*). A positional inverted index supports keyword (TF-IDF ranked), phrase and Boolean (AND/OR/NOT, brackets) search. On 14 domain queries with relevance judgments fixed **before** retrieval, the Final Pipeline reaches macro precision 0.43, recall 0.94, F1 0.53 and P@3 0.52. That matches the best alternative pipeline and beats the stemming pipeline, while producing a readable, smaller index with far more meaningful n-grams. A Gradio GUI demonstrates every stage live.

1. Domain and motivation

People looking for mental-health information read guidance written for two audiences: **clinical guidelines** (NICE, WHO mhGAP) and **public information** (NIMH, NHS, WHO fact sheets). This text has properties that general-purpose NLP handles badly, and errors here can matter:

Domain property	Example	Why default NLP fails
Negation carries the advice	"Do not stop taking an antidepressant suddenly"	Standard stop lists delete <i>not</i> , which reverses the advice
Modal strength	"should not be used", "may", "must"	Removed as stop words; the strength of a recommendation is lost
Hyphenated and coded terms	self-harm, 12-week, DSM-IV, ICD-11, 1-800-662-4357	Split into fragments by default tokenizers
Acronyms and plural acronyms	GAD, PTSD, CBT, SSRIs	Tagged as organisations; <i>SSRIs</i> not merged with <i>SSRI</i>
Drug and therapy names	sertraline, naltrexone, psychotherapy	No NER label in general models
Scaffolding numbers	[2004, amended 2020], 1.2.30, p-values	Noise terms in the index
PDF layout artefacts	running headers/footers, bullets, broken lines	Fake tokens and fake n-grams

The system design follows the AS1 sequence: **Select** → **Order** → **Implement** → **Compare** → **Evaluate** → **Justify**.

2. Corpus

2.1 Documents

ID	Title	Organisation	Layer	Audience	Format	Words
D01	Depression in adults: treatment and management (sections)	NICE (UK)	A	professional	PDF	4,278
D02	Generalised anxiety disorder and panic disorder in adults: management (sections)	NICE (UK)	A	professional	PDF	3,029
D03	Self-harm: assessment, management and preventing recurrence (sections)	NICE (UK)	A	professional	PDF	1,582
D04	mhGAP guideline, 3rd ed.: depression and anxiety modules (pp. 50–59, 90–95)	WHO	A	professional	PDF	6,813
D06	Depressive disorder (depression) fact sheet	WHO	B	public	HTML	1,197
D07	Depression	NIMH (US)	B	public	PDF	2,601
D08	Generalized Anxiety Disorder: What You Need to Know	NIMH (US)	B	public	PDF	1,961
D09	Post-Traumatic Stress Disorder	NIMH (US)	B	public	PDF	2,162
D10	Eating Disorders: What You Need to Know	NIMH (US)	B	public	PDF	1,497
D11	Frequently Asked Questions About Suicide	NIMH (US)	B	public	PDF	1,827
D12	Raising low self-esteem	NHS (UK)	B	public	HTML	967
D13	Get help with loneliness	NHS (UK)	B	public	HTML	892
D14	Antidepressants (incl. stopping or coming off)	NHS (UK)	B	public	HTML	1,441
D16	I'm So Stressed Out! Fact Sheet	NIMH (US)	B	public	PDF	890
D17	Sleep problems (Every Mind Matters)	NHS (UK)	B	public	HTML	1,154
D24	mhGAP guideline 2023: Alcohol use disorders module (pp. 44–49)	WHO	A	professional	PDF	2,686
D25	mhGAP guideline 2023: Drug use disorders module (pp. 96–107)	WHO	A	professional	PDF	6,320
	Total					41,297

Layer A = clinical guidelines; *Layer B* = public information. **Licences** are recorded per document in `data/metadata/source_manifest.csv`: NIMH is US public domain; NHS is the Open Government Licence; WHO is CC BY-NC-SA 3.0 IGO (guidelines) or WHO copyright with non-commercial reuse (fact sheet); NICE is © NICE, used for academic purposes and not redistributed.

Gaps (recorded, not hidden): a planned *Layer C* (Indian helpline information, Tele-MANAS, D20) is empty, because the only available file was an image-only PDF that failed quality control. Two planned documents (D05, D19) could not be obtained. D24 and D25 were taken from the larger WHO mhGAP PDF to cover substance-use topics the other documents did not. The corpus contains **no medication doses** (Exercise 6 found none), so dose handling could not be tested.

2.2 Corpus construction pipeline (`src/build_corpus.py`)

register → extract → clean → section → QC → build → validate

- **Extraction:** PyMuPDF for PDFs (page-aware), BeautifulSoup for HTML (navigation, scripts, video players and photo credits removed).
- **Cleaning (ordered rules):** remove repeated header/footer lines (the top and bottom 5 lines of each page are compared across pages), normalise bullets to one marker, unwrap PDF line breaks (never joining a bullet item to the previous line), remove boilerplate repeated on $\geq 40\%$ of pages, normalise whitespace. Case, punctuation, stop words and numbers are **kept**, so the NLP experiments see real text.

- **Sectioning:** only the guideline sections relevant to the domain (`pages:` / `toc:` selectors in the manifest).
- **Quality control:** thresholds for length, non-alphabetic ratio, dictionary-word ratio, empty pages, URL and citation density, and near-duplicates (5-word shingles). D20 was rejected (image-only).
- **Output:** `data/processed/corpus.json` (text + metadata for every document); final validation reported 0 errors.

Cleaning defects found through statistics and fixed (with dry runs before applying):

1. The NICE 5-line page footer survived a 3-line header/footer zone, so the zone was widened to 5 lines.
2. Embedded video-player text and photo credits in NHS/WHO HTML were removed.
3. Bullet items were merged into run-on lines, and WHO's symbol-font bullet appeared as a stray "y". Both are now handled with a bullet marker kept until unwrapping.
4. A WHO running header on alternate pages slipped under the 50% boilerplate threshold, which was lowered to 40%.

A remaining minor defect is a 3-page print header in D16 (below the 4-page minimum for boilerplate detection).

3. Module 1: Document statistics

Per-document sentences, tokens, characters and vocabulary are in `results/document_statistics.csv`. On whitespace tokens, the corpus has **2,331 sentences, 39,961 tokens, 271,378 characters and a vocabulary of 5,421**. The two longest documents are D04 and D25, both WHO guideline excerpts.

Table A: Document and preprocessing results (`preprocessing_results.csv`)

Measure	Before processing	After processing (Final Pipeline)	Change
Documents	17	17	0%
Tokens	39,961	24,913	-37.7%
Unique tokens	5,976	2,919	-51.2%
Vocabulary size	5,421	2,919	-46.2%
Average tokens per document	2,350.6	1,465.5	-37.7%

"Before" = whitespace tokens of the cleaned text. "After" = terms produced by the Final Pipeline (Section 5). Vocabulary roughly halves, while negation, modal verbs and domain terms are kept.

4. Module 2: Exercises 1–14

Each exercise compared the methods on this corpus, measured the results, and ended in a decision. Labels: **(A)** required by AS1; **(B)** adapted to the domain; **(C)** our own design choice.

Exercise 1: Dictionary and term dictionary

Tokenizer	Tokens	Dictionary size	Term dictionary size	Reduction
<code>split()</code>	39,961	5,976	3,804	36.3%
NLTK <code>word_tokenize</code>	46,218	4,356	3,743	14.1%
spaCy tokenizer	47,429	4,254	3,654	14.1%

`split()` inflates the dictionary with punctuation attached to words (2,170 entries such as "depression," and "(CBT)"). The *term dictionary* (lowercased, punctuation stripped) is the meaningful size. The categories of dictionary entries (plain words, numbers, hyphenated words, URLs, case duplicates) are in `ex2_dictionary_categories.csv`.

Exercise 2: Sentence splitting

Splitter	Units	Avg words	Long units (> 60 words)	Fragments (< 3 words)	Time (s)
<code>split(". ")</code>	2,340	17.3	52	350	0.0
NLTK <code>sent_tokenize</code>	1,943	20.6	62	33	0.0
spaCy <code>sentencizer</code>	1,975	20.3	53	35	0.7
spaCy <code>en_core_web_sm</code>	1,902	21.0	60	19	18.8
LineTokenizer + <code>sent_tokenize</code>	3,337	12.0	5	601	0.1

Guidance text is full of headings and bullet lists without final full stops. Sentence-only splitters merge them into long run-on "sentences". **Decision (B):** split into lines first, then sentences. Run-on units fall from 62 to 5. The extra "fragments" are real headings and bullet items, which is the correct unit for n-grams, POS tagging and NER.

Exercise 3: Tokenization (Table B)

We listed ten domain tokenization problems (P1–P10) and measured how many corpus occurrences each tokenizer keeps intact.

Problem	Example	NLTK	spaCy	Custom	Hybrid
P1 hyphenated compound	self-harm	96.2%	0%	98.0%	98.0%
P3 numeric range	6–12	0%	100%	100%	100%
P5 classification code	DSM-IV, ICD-11	91.9%	87.1%	100%	100%
P6 URL	https://findtreatment.gov	68.3%	95.2%	100%	95.2%
P7 helpline number	1-800-662-4357	100%	0%	100%	100%
P9 slash term	discontinuation/withdrawal	87.5%	44.4%	93.1%	93.1%
P10 abbreviation	e.g.	0%	97.6%	100%	97.6%
Negation	doesn't → <i>does not</i>	n't kept (14)	n't kept (14)	not (0 n't)	not
Time (s)		0.34	0.66	0.18	0.60

Table B (`tokenization_comparison.csv`) shows one real corpus sentence for each problem under all four tokenizers. Examples: spaCy splits *self-harm* into *self - harm* and the helpline number into seven pieces; NLTK splits *e.g.* and the URL; the custom tokenizer gets all ten right.

Custom tokenizer (C): a pre-step normalises the non-breaking hyphen (U+2011) and expands negative contractions (*can't* → *can not*, *n't* → *not*). Then an NLTK `RegexTokenizer` applies ordered rules: URLs → helpline numbers → recommendation IDs → ranges → codes → number-word compounds → hyphenated compounds → slash terms → abbreviations → numbers → words → symbols. **Hybrid** = spaCy's tokenizer with the corpus's domain terms added as special cases; it is used only where a spaCy model must tokenize (POS, NER). **Decision:** the Custom tokenizer is used for documents and queries. It is the most accurate on domain tokens and the fastest.

Exercise 4: Byte-Pair Encoding (Table G)

BPE was trained with Hugging Face `tokenizers` (BPE model, whitespace pre-tokenizer, `BpeTrainer`).

Target vocabulary	Learned	Corpus tokens	Tokens per word
500	500	96,247	2.41
1,000	1,000	75,668	1.89
2,000	2,000	60,843	1.52
4,000	4,000	51,985	1.30
8,000	6,913	48,496	1.21

Table G (`bpe_results.csv`) shows how each word is split. Frequent words stay whole, but rare domain terms are split at small vocabulary sizes: *benzodiazepines* becomes 9 pieces at vocabulary 500, *psychotherapy* becomes *psych* | *other* | *apy*, *PTSD* becomes *P* | *T* | *S* | *D*, and *self-harm* becomes *self* | - | *harm* at every size. **Decision:** BPE is analysed but **not used** in the pipeline. Sub-word pieces are not meaningful search terms, and a whole-word index fits a 17-document corpus. BPE would matter for a neural model with a fixed vocabulary.

Exercise 5: Phrase structure

spaCy noun chunks and dependency-based verb phrases show that the corpus is **instructional**: verb phrases carry the advice (*seek help*, *talk to your doctor*, *not stop taking*), and many depend on negation (*not go away with treatment*, *not smoke*). Top noun phrases per document identify each document's topic (e.g. *panic disorder*, *psychological interventions*, *ptsd*, *low self-esteem*). This motivated keeping negation and phrase information (`ex5_instruction_phrases.csv`, `ex5_top_noun_phrases.csv`).

Exercise 6: Numbers

Category	Occurrences	Policy
Duration (6 months, 12 weeks)	72	keep
Age, sessions/frequency	8 + 8	keep
Percentage	35	keep
Helpline number (988, 111, 999, 1-800-...)	28	keep
Dose	0	(keep; none in the corpus)
NICE year tag [2004, amended 2020]	56	drop
Calendar date, year	4 + 168	drop
Recommendation ID (1.2.30)	73	drop
Statistical value (0.31)	23	drop

Decision (C): a number policy runs before tokenization. It drops scaffolding numbers (number tokens 886 → 613, vocabulary 4,367 → 4,255) and keeps every category a user might search for. A check confirmed that all kept categories are unchanged.

Exercise 7: Stop words

Stop list	Size	Tokens kept	Vocabulary	Negation tokens kept	Modal tokens kept
None	0	39,660	3,662	327	828
NLTK	198	24,570	3,539	22	326
spaCy	326	22,968	3,438	0	0
Custom	190	24,913	3,546	327	828

A meaning test on real sentences (`ex7_meaning_test.csv`) shows the problem. With the standard lists, "Antidepressant medicines **should not** be used for treating depression in children" becomes *antidepressant*

medicines used treating depression children, which reverses the advice. **Custom list (B)**: (NLTK list – negation, modal, timing and amount words) + generic corpus words (*people, person, things, information, time, et, al*). It removes almost as many tokens as NLTK and keeps every negation and modal. Word clouds before and after are in the notebook.

Exercise 8: Stemming

Stemmer	Words → stems	Reduction	Problem seen
Porter	3,546 → 2,571	27.5%	non-words (<i>activ</i>)
Snowball	3,546 → 2,545	28.2%	non-words (<i>anxieti, depress</i>)
Lancaster	3,546 → 2,358	33.5%	over-merging: <i>car/care, us/use</i>
Regexp (suffix rules)	3,546 → 2,994	15.6%	misses irregular forms
Domain light stemmer (custom)	3,546 → 3,025	14.7%	safe but weak

Snowball is the best stemmer, but every stemmer produces non-words or harmful merges (*medication* and *medical* both become *medic*). Stemming became Pipeline A in Module 3.

Exercise 9: Order of stop-word removal and stemming

Order	Tokens	Vocabulary
Stop-word removal → stemming	24,913	2,545
Stemming → stop-word removal	25,544	2,553

Stemming first changes surface forms, so 21 NLTK stop words no longer match the list and survive into the index (*any* → *ani*, *because* → *becaus*, *does* → *doe*, *very* → *veri*). **Decision**: stop words are matched on the **surface form** before or independently of normalisation.

Exercise 10: Lemmatization (Table C)

Method	Vocabulary	Reduction	Time (s)
None	3,546	n/a	0.0
Snowball stemmer	2,545	28.2%	0.3
WordNet (no POS)	3,248	8.4%	0.2
WordNet + POS	2,966	16.4%	2.3
TextBlob (+ POS for verbs)	2,980	16.0%	2.9
spaCy (in context)	2,926	17.5%	13.9

Table C (`stem_vs_lemma.csv`) compares forms word by word: *antidepressants* → stem *antidepress* vs lemma *antidepressant*; *anxiety* → *anxieti* vs *anxiety*; *slept* → *slept* vs *sleep*; *medication* → *medic* vs *medication*. **Decision**: spaCy lemmatization, running spaCy's tagger on *our* tokens (no re-tokenization) so lemmas use sentence context. Domain rules were added through spaCy's `attribute_ruler`: plural acronyms map to the singular (SSRIs → *ssri*, RCTs → *rct*), *media* stays *media*, and *generalised/generalized* → *generalised*. Gensim was excluded because its lemmatizer was removed in version 4.

Exercise 11: Combined preprocessing

The decisions above were combined into one function, `final_preprocess` (the Final Pipeline), and its effect measured: Table A (Section 3).

Exercise 12: POS tagging and custom POS taggers (Table D)

Gold standard. 50 seeded sentences (seed 134) containing domain terms, 724 tokens. Tokens where the NLTK and spaCy taggers agree were accepted. All 119 disagreements and domain tokens were reviewed by hand using Penn Treebank conventions (proposed by the AI assistant, approved by the team). The gold set was split **by sentence**: 30 for ML training, 20 held out (303 tokens, 23 domain tokens). Every tagger is scored on the same held-out sentences. spaCy's tags were mapped to NLTK's conventions for brackets and hyphens.

Domain terms mis-tagged by the default taggers include imperatives at the start of a sentence (*Follow, Learn*: NLTK tags them as nouns), plural acronyms (*SSRIs, RCTs*: NLTK says NNP, gold NNS), *e.g./i.e.* (NLTK: VB/JJ/NN, gold FW), *panic* in "panic disorder" (NLTK: JJ), and *NIMH/PTSD* (spaCy: JJ/NN).

Custom taggers

- **Rule-based (dictionary + regex):** NLTK tags, overridden by a domain lexicon (acronyms, medicines, *e.g.* → FW) and regex rules (ranges and helpline numbers → CD, codes → NNP, plural acronyms → NNS, web addresses → NNP).
- **ML 1: n-gram backoff:** Trigram → Bigram → Unigram → Affix → Default(NN), trained on the Penn Treebank sample, with and without the 30 domain sentences.
- **ML 2: KNN:** features are the word, previous word, next word, word length, and first and last character (one-hot encoded), trained on Treebank + domain sentences.

Tagger	Accuracy (%)	Domain-token accuracy (%)
Default: NLTK perceptron	92.4	65.2
Default: spaCy	93.4	60.9
Custom: rule-based (dictionary + regex)	93.7	82.6
ML: n-gram backoff (Treebank only)	80.2	34.8
ML: n-gram backoff (Treebank + 30 domain sentences)	88.4	82.6
ML: KNN, K = 3 (Treebank + domain)	87.1	82.6

KNN by K: K=1: 86.5, **K=3: 87.1**, K=5: 85.8, K=7: 85.8. With context and word identity removed (shape features only), accuracy falls to 71.3.

Findings. The rule-based tagger improves domain-token accuracy by **+17.4 points** over the best default tagger, and is also best overall. Its remaining domain errors are web addresses (our rule says NNP; the gold follows the NN convention, and we did not change the rule after seeing the test data) and "ways to **self-harm**", a verb use that a context-free dictionary cannot see. For the ML taggers, **domain training data matters more than the algorithm**: adding only 30 domain sentences gives the largest improvement in the table. **Decision (C)**: the rule-based tagger is our custom POS tagger; it is accurate, transparent and needs no annotated data. Table D (`pos_tagging_results.csv`) lists every held-out domain token with its default tags, custom tag, gold tag and Correct/Incorrect. *Limitation*: a small test set (23 domain tokens), and tokens where both taggers agree were not individually reviewed.

Exercise 13: Named entity recognition (Table E)

Applying default NER (corpus counts in `ex13_entity_counts.csv`). In this corpus **MONEY and EVENT never occur, and PRODUCT almost never does**. DATE is mostly publication years in citations; PERSON is mostly citation authors. spaCy labels domain acronyms as ORG (GAD 63×, CBT 60×). NLTK `ne_chunk` labels capitalised ordinary words as GPE (*Key, Research, Further, Talk*). Of the domain-entity mentions found by our rules, the default spaCy model **missed 960 completely** and gave **189 the label ORG** (`ex13_domain_entities.csv`): *depression* missed 179×, *antidepressants* 76×, 988 tagged as CARDINAL.

Custom NER (C): a spaCy `EntityRuler` placed before the statistical NER, with domain labels **CONDITION**, **MEDICATION**, **THERAPY**, **HELPLINE** and organisation acronyms (NICE, NHS, WHO, NIMH, SAMHSA). Patterns are case-sensitive where the lower-case form is an ordinary word (*who*, *nice*). The helpline pattern is regex-based (988, 1-8xx numbers).

Evaluation. 40 seeded sentences, 74 gold entities (proposed by the AI assistant, approved by the team), exact span + label match. Numbers such as citation markers (CARDINAL) are out of scope.

Scope	System	Predicted	Correct	Incorrect	Missed	Precision	Recall	F1
Required types	spaCy (default)	32	17	15	12	0.53	0.59	0.56
Required types	NLTK <code>ne_chunk</code>	33	8	25	21	0.24	0.28	0.26
Required types	spaCy + EntityRuler	28	18	10	11	0.64	0.62	0.63
Required + domain	spaCy (default)	32	17	15	57	0.53	0.23	0.32
Required + domain	NLTK <code>ne_chunk</code>	33	8	25	66	0.24	0.11	0.15
Required + domain	spaCy + EntityRuler	64	51	13	23	0.80	0.69	0.74

Entity-level Table E (`ner_entities.csv`) lists each entity with its predicted type and Correct / Incorrect / Missed. The remaining errors of the custom system are (i) errors inherited from spaCy on citation names ("Chan et al." span, *Buchholz* tagged as GPE), and (ii) dictionary limits: unlisted drugs (fluvoxamine, amitriptyline, TCAs), plural forms (*eating disorders*), span boundaries (*Major depression*) and context (*people who self-harm* is a verb).

Decision: spaCy + domain EntityRuler, which is also used in the GUI.

Exercise 14: N-gram analysis (Table F)

N-grams are built **inside** sentences and lines, on content tokens with stop words kept (so phrases stay readable).

N-gram	Total	Unique	Seen once	Top meaningful phrases (no stop word at either end)
Unigram	39,660	3,662	40.6%	treatment, health, interventions, depression, symptoms
Bigram	36,454	16,632	67.0%	mental health, psychological interventions, health care, panic disorder, side effects, drug use
Trigram	33,501	23,079	80.2%	health care provider, certainty of evidence, drug use disorders, strength of recommendation
4-gram	30,774	23,852	85.4%	adults with anxiety disorders, side effects of medication, cognitive behavioural therapy cbt
5-gram	28,213	22,886	87.6%	potential side effects of medication, national institute of mental health

The raw top-10 lists are dominated by function words (*of the*, *should be*). The check *number of n-grams* = $N - n + 1$ per sentence holds for every n. Frequent 4- and 5-grams are largely **repeated template text** from NICE tables ("suit people who do not like talking about their depression").

Dictionary size vs term dictionary: term dictionary 2,919; unigram 3,662 (1.3×); bigram 16,632 (5.7×); trigram 23,079 (7.9×); 4-gram 23,852 (8.2×); 5-gram 22,886 (7.8×), and the longer dictionaries consist almost entirely of one-off entries. **Decision (C):** do **not** index n-grams. The index stores single terms with **positions**, so phrase queries are answered by checking adjacent positions.

5. Module 3: Process selection and ordering

5.1 The Final Pipeline and why each step is where it is

#	Process	Required?	Position	Depends on	If moved
1	Extraction	Yes: PDF/HTML are not text	first	raw files	nothing can run
2	Cleaning	Yes: layout artefacts create fake tokens	after extraction	page and line structure	after tokenization, headers are already tokens
3	Sentence splitting (lines first)	Yes, for n-grams, POS, NER and phrase boundaries	after cleaning	bullets on their own lines	before unwrapping, every PDF line is a "sentence"
4	Number policy	Yes: scaffolding numbers are noise	before tokenization	raw patterns ("1.2.3", "[2022]")	after tokenization the patterns are already split
5	Custom tokenization	Yes	after number policy	cleaned text	everything later depends on it
6	Lowercasing	Yes	after tokenization	tokens	before, case-based rules (acronyms) are lost
7	Punctuation removal	Yes	after lowercasing	tokens	before tokenization it breaks self-harm, e.g., URLs
8	Lemmatization (spaCy, in context, domain rules)	Yes	before stop-word removal	the full sentence for POS context	after stop-word removal, lemmas change (tested below)
9	Stop-word removal (custom list)	Yes	after lemmatization, by surface form	the list keeps negation and modals	the standard list deletes <i>not</i> ; before lemmatization, see row 8
—	Stemming	No (non-words, over-merging)	(alternative to row 8)	n/a	tested as Pipeline A
—	BPE	No (splits domain terms)	not used	n/a	n/a
—	POS / NER	Analysis branch (reports, GUI)	parallel, on original-case sentences	case and context	after lowercasing or stop-word removal they lose their evidence
—	N-grams	Analysis; positions replace them	after normalisation	sentence boundaries	across boundaries they create fake phrases
10	Positional inverted index	Yes	after the pipeline	final terms + positions	built on raw tokens, documents and queries would not match
11	Query processing	Yes: the same function as for documents	query time	the pipeline	a different pipeline means <i>antidepressants</i> ≠ <i>antidepressant</i>
12	Retrieval, ranking, evaluation	Yes	last	index + judgments made before retrieval	n/a

5.2 Mandatory experiment: pipeline orders

All pipelines share steps 1–7 and differ only in the branch of the AS1 diagram and its order. They run per sentence on the same corpus.

Pipeline	Steps	Tokens	Vocabulary	Readable vocabulary	Meaningful top-20 bigrams	Negation bigrams	Domain terms readable (of 14)	Time (s)
A	NLTK stop words → Snowball stemming	24,570	2,535	56.3%	7	26	9	0.6
B	no stop-word removal → lemmatization	39,660	3,011	89.5%	5	320	13	29.2
B-reordered	custom stop words → lemmatization	24,913	2,940	90.5%	17	320	13	27.4
Final	lemmatization → custom stop words	24,913	2,914	90.5%	17	320	13	30.1

"Readable" = a WordNet word, a known domain term, or a kept number. The one "unreadable" domain term in B and Final is *should*, which is not a WordNet entry.

What each pipeline does to domain terms (`module3_domain_terms.csv`):

Input	Pipeline A	Pipeline B	Final
generalised anxiety disorder	generalis anxiety disord	generalised anxiety disorder	generalised anxiety disorder
antidepressants	antidepress	antidepressant	antidepressant
SSRIs	ssris	ssri	ssri
do not stop taking	stop take	do not stop take	not stop take
should not be used	use	should not be use	should not use
talking therapies	talk therapi	talk therapy	talk therapy

- **Pipeline A** has the smallest vocabulary, but almost half of it is not words. It turns "should not be used" into *use*, which reverses safety advice.
- **Pipeline B** keeps everything: 60% more tokens, and function-word bigrams dominate (*do not, there be*).
- **Order matters:** with identical steps, removing stop words *before* lemmatization changed lemmas in **231 of 3,337 sentences**. Inflections stay unmerged (*outcomes, tapering, discontinuities*), giving a larger vocabulary. Lemmatization therefore precedes stop-word removal. The experiment also exposed one spaCy error: *self-harmed* → *self-harme*. It can be fixed with a lemma rule (recorded as future work).

Table H: Pipeline comparison (`pipeline_comparison.csv`)

Measure	Pipeline A	Pipeline B	Final Pipeline
Token count	24,570	39,660	24,913
Vocabulary size	2,535	3,011	2,914
Meaningful n-grams (of top-20 bigrams)	7	5	17
Precision	0.414	0.429	0.429
Recall	0.896	0.943	0.943
F1-score	0.505	0.534	0.534

(Precision, recall and F1 come from Module 6.)

6. Module 4: Information retrieval

Index (C): a positional inverted index (term → document → positions), saved as `results/inverted_index.json`. Positions count Final Pipeline terms, and they jump at every sentence boundary so a phrase cannot match across sentences or headings. Posting lists in the AS1 format:

```
depression -> D01, D02, D03, D04, D06, D07, D08, D09, D10, D11, D12, D13, D14, D16, D25
ssri       -> D01, D02, D04, D06, D08, D09, D14
self-harm  -> D02, D03, D04, D06, D14
naltrexone -> D24, D25
988        -> D07, D08, D09, D10, D11, D16
```

Index statistic	Value
Documents	17
Vocabulary (terms)	2,914
Postings (term–document pairs)	8,125
Positions stored	24,913
Average documents per term	2.79
Terms in only one document	51.4%
Terms in every document	10 (<i>also, can, health, help, may, mental, more, need, not, social</i>)

The index vocabulary (2,914) is slightly smaller than Table A's (2,919) because the index lemmatizes sentence by sentence; Table A lemmatized whole documents, which gives spaCy slightly different context.

Search modes

- **Keyword:** documents containing *any* query term, ranked by log-scaled TF × IDF: $\text{score}(d) = \sum (1 + \log \text{tf}) \cdot \log(N/\text{df})$.
- **Phrase:** query terms at consecutive positions.
- **Boolean:** AND, OR, NOT in **capitals** (lower-case *not* is a search term, because negation is kept), quotes for phrases, brackets, precedence NOT > AND > OR (shunting-yard parser). Adjacent words form one operand (implicit AND), and "x NOT y" means "x AND NOT y". The parser was tested on 14 cases including malformed queries.
- Every query passes through the **same** `pipeline()` as the documents.

Examples: "do not stop" matches **all 17** documents as keywords but **1** document (D14) as a phrase. The keyword query "sleep problems" ranks the NHS sleep page (D17) first among 16 matches.

7. Module 5: Domain-specific queries (Table I)

Fourteen queries cover every type in the AS1 examples, plus natural-language queries. The queries and relevance judgments (Section 8) were written **before** any retrieval was run on them, and the Module 4 demo queries were not reused.

ID	Query	Query type	Retrieved documents	# Results	Time (ms)
Q01	anxiety	Unigram (keyword)	D04, D08, D16, D02, D13, D07, D10, D14, D01, D17, D03, D09, D12	13	6.6
Q02	panic disorder	Bigram (phrase)	D04, D02, D03, D07, D09, D14	6	6.8

ID	Query	Query type	Retrieved documents	# Results	Time (ms)
Q03	generalised anxiety disorder	Trigram (phrase)	D08, D04, D02, D03, D07, D14	6	5.8
Q04	antidepressant side effects	Trigram (keyword)	D14, D01, D02, D07, D04, D08, ...	14	4.7
Q05	how to stop taking antidepressants safely	4-gram (keyword, natural language)	D14, D01, D04, D02, ...	17	6.5
Q06	alcohol AND medication	Boolean AND	D01, D07, D02, D08, D09, D24, D14	7	11.8
Q07	suicide OR self-harm	Boolean OR	D03, D11, D06, D04, D07, ...	12	13.5
Q08	depression AND NOT anxiety	Boolean NOT	D06, D11, D25	3	12.0
Q09	trouble sleeping	Bigram (keyword)	D08, D11, D07, D09, D17, ...	12	6.8
Q10	PTSD OR trauma	Boolean OR	D09, D14, D25	3	13.8
Q11	crisis helpline	Bigram (keyword, natural language)	D11, D03, D07, D06, D08, D09, D10, D14, D16	9	17.0
Q12	low self-esteem	Bigram (phrase)	D12, D07	2	4.4
Q13	loneliness	Unigram (keyword)	D13	1	5.2
Q14	"drug use disorders" AND psychosocial	Combined (phrase + AND)	D25	1	14.0

Q03 confirms that the domain lemma rule works: the US spelling *generalized* in D08 matches the query *generalised*, and D08 is ranked first. Boolean result sets are exact; they are ordered by the TF-IDF score of the query words so that P@K is defined for every query type.

8. Module 6: Retrieval evaluation

Relevance set. For each query we wrote an information need and judged **every query × document pair** (14 × 17, `relevance_judgments.csv`). A document is relevant if it gives **substantive** information answering the need; passing mentions, citations, cross-references and generic advice ("talk to your doctor") are not relevant. Judgments were proposed by the AI assistant after reading the relevant passages of each document, and reviewed and approved by the team.

Per-query results, Final Pipeline (`evaluation_per_query.csv`)

ID	Type	Retrieved	Relevant	P	R	F1	P@3	R@3	P@5	R@5
Q01	Keyword	13	4	0.31	1.00	0.47	1.00	0.75	0.80	1.00
Q02	Phrase	6	2	0.33	1.00	0.50	0.67	1.00	0.40	1.00
Q03	Phrase	6	3	0.50	1.00	0.67	1.00	1.00	0.60	1.00
Q04	Keyword	14	5	0.36	1.00	0.53	1.00	0.60	0.80	0.80
Q05	Keyword (NL)	17	3	0.18	1.00	0.30	0.67	0.67	0.60	1.00
Q06	Boolean AND	7	1	0.14	1.00	0.25	0.00	0.00	0.00	0.00
Q07	Boolean OR	12	2	0.17	1.00	0.29	0.67	1.00	0.40	1.00
Q08	Boolean NOT	3	3	0.33	0.33	0.33	0.33	0.33	0.20	0.33
Q09	Keyword	12	1	0.08	1.00	0.15	0.00	0.00	0.20	1.00
Q10	Boolean OR	3	1	0.33	1.00	0.50	0.33	1.00	0.20	1.00
Q11	Keyword (NL)	9	8	0.78	0.88	0.82	0.67	0.25	0.60	0.38
Q12	Phrase	2	1	0.50	1.00	0.67	0.33	1.00	0.20	1.00

ID	Type	Retrieved	Relevant	P	R	F1	P@3	R@3	P@5	R@5
Q13	Keyword	1	1	1.00	1.00	1.00	0.33	1.00	0.20	1.00
Q14	Combined	1	1	1.00	1.00	1.00	0.33	1.00	0.20	1.00
	Macro average			0.43	0.94	0.53	0.52	0.69	0.39	0.82

P@K divides by K even when fewer than K documents are retrieved (standard definition), which penalises single-result queries.

Table J: Overall performance (`evaluation_results.csv`)

Each pipeline has its **own index**; queries are processed by the same pipeline; all use the same judgments.

Method / Pipeline	Precision	Recall	F1-score	P@3	R@3	P@5	R@5	Indexing time (s)	Avg query time (ms)
Pipeline A (stop words → stemming)	0.414	0.896	0.505	0.452	0.665	0.371	0.846	0.6	0.9
Pipeline B (no stop-word removal → lemmatization)	0.429	0.943	0.534	0.524	0.686	0.386	0.822	29.2	8.0
Final Pipeline (lemmatization → custom stop words)	0.429	0.943	0.534	0.524	0.686	0.386	0.822	30.1	7.8

8.1 Analysis

- **High recall, low precision.** Keyword search returns documents containing *any* query term, and domain words occur widely across a focused corpus. Almost every relevant document is found (recall 0.94), along with many others. **Ranking does work:** P@3 (0.52) is higher than overall precision (0.43).
- **Phrase and Boolean queries are stricter.** Q12–Q14 retrieve one or two documents with high precision.
- **Error analysis** (`module6_error_analysis.csv`):
 - Q09 "*trouble sleeping*": the sleep page D17 is ranked 5th. *Trouble* is a rare, high-IDF word, so documents with "trouble concentrating" (D08) or "legal troubles" (D11) outrank the page about sleep, which is a TF-IDF weakness on short queries.
 - Q06 "*alcohol AND medication*": the relevant alcohol module D24 is found, but ranked 6th of 7. Both words co-occur in many documents (e.g. "avoid alcohol with medication"), so a Boolean AND does not express the need *medicines for alcohol use disorder*.
 - Q08 "*depression AND NOT anxiety*": **NOT removes relevant documents** (D01 and D07 are about depression but mention anxiety) and keeps documents that merely mention depression (D11, D25). Boolean NOT is a blunt tool on documents that discuss related conditions.
 - Q11 "*crisis helpline*": *helpline* does not occur in the documents ("Lifeline", "crisis line"). Results depend on *crisis*, and D13 (NHS 111 "urgent help") is missed: **vocabulary mismatch**.
 - Q02/Q03 *phrases*: D03 matches only because it cites another guideline's title: a correct phrase match but not a relevant document.
- **Pipeline A** is lowest on every metric: stems conflate or separate forms unpredictably and negation is lost.
- **Pipeline B and the Final Pipeline score identically.** The words B keeps (*to, how, of, the*) occur in nearly every document, so their IDF is ~0: they change neither the ranking nor the set found. With 17 documents and document-level judgments, retrieval metrics **cannot separate** these two pipelines.

9. Final selection and justification

Selected: the Final Pipeline number policy → custom tokenizer → lowercase → punctuation removal → spaCy lemmas in context (+ domain rules) → custom stop words (surface form)

1. **Retrieval effectiveness:** equal to the best alternative (Pipeline B) and better than stemming (Pipeline A) on precision, recall, F1 and P@3 (Tables H and J).
2. **Index size:** 37% fewer tokens than Pipeline B for the same retrieval quality; vocabulary 2,914 vs 3,011.
3. **Readable terms and phrases:** 90.5% readable vocabulary (vs 56.3% for stemming) and 17 of the top-20 bigrams are meaningful (vs 7 and 5). This matters because terms and phrases are **shown to users** in the GUI and the dictionary.
4. **Safety-relevant meaning is preserved:** negation and modal strength survive ("not stop take", "should not use"). This is the main domain risk, even where document-level metrics cannot measure it.
5. **Cost:** slower than stemming (about 30 s to index instead of under 1 s), but that is paid once; a query takes about 8 ms.

10. GUI (Gradio): `python app.py`

The GUI follows the suggested layout and demonstrates the complete working pipeline, using the same code (`src/nlp_pipeline.py`), the saved index and the results CSVs.

Tab	Content
1 · Documents	Load Documents: document statistics and the document-ID → name table
2 · Text Analysis	Live, on any typed text: tokenization (NLTK vs spaCy vs custom), preprocessing (stage-by-stage trace of the Final Pipeline), POS (default vs custom, with changed tags marked), NER (highlighted domain entities vs default spaCy), n-grams (n = 1–5), BPE ; each with the saved table (A–G) underneath
3 · Terms Dictionary	The inverted index as a table (term, documents with counts, occurrences, document frequency), sortable by term or document frequency, with a filter
4 · Search & Evaluation	Query, query type (Keyword / Phrase / Boolean), SEARCH ; retrieved documents, number of results, execution time, Precision, Recall, F1, P@3 (live for the 14 judged queries, selectable from a list); results marked relevant / not relevant, with snippets and source links
5 · Compare Pipelines	The same query on Pipeline A and the Final Pipeline side by side, each with its own metrics, plus Tables H and J
6 · Results Tables	All CSV checkpoints (Tables A–J and supporting tables)

A "not medical advice" note with crisis numbers (988 US; 999/111 UK) is shown on every page. Example demo: *Compare Pipelines* → "do not stop taking antidepressants" (Phrase). Pipeline A, which has dropped *not*, returns "...if you want to **stop taking** an antidepressant"; the Final Pipeline returns "Do **not stop taking** an antidepressant suddenly..."

11. Limitations and future work

- **Small evaluation:** 17 documents, 14 queries, binary judgments by one team; POS and NER gold sets are small (23 held-out domain tokens; 74 entities). Gold labels were proposed by an AI assistant and approved by the team, not annotated independently by several people.
- **Document-level retrieval** cannot separate pipelines that differ only in function words. **Passage-level** retrieval (sections or sentences) would give sharper rankings and evaluation.
- **TF-IDF weaknesses:** rare-word dominance (Q09), vocabulary mismatch (Q11). Future work: BM25, query expansion with a domain synonym list (helpline ↔ lifeline ↔ crisis line), or dense embeddings.
- **Dictionary-based components** (custom POS lexicon, EntityRuler) miss unlisted terms (fluvoxamine, amitriptyline) and plural patterns; they could be extended automatically from drug lists, or replaced by a trained NER with domain data.

- **Corpus gaps:** no Indian layer (image-only Tele-MANAS PDF), no dose information; a small residual print header in D16.
- **Known lemma error:** *self-harmed* → *self-harme* (spaCy); fix with a domain lemma rule.

12. Reproducibility

- **Environment:** Python 3.14; `pip install -r requirements.txt`; `python -m spacy download en_core_web_sm`.
- **Corpus:** `python src/build_corpus.py`, which writes `data/processed/corpus.json` from `data/raw/` and `data/metadata/source_manifest.csv`.
- **Notebook:** run `Domain_Text_Analysis_Retrieval.ipynb` top to bottom; every table is saved to `results/`. All randomness uses **SEED = 134**.
- **GUI:** `python app.py`, then open the printed local URL.
- **Folder structure**

```
Domain_Text_Analysis_Retrieval/
├── Domain_Text_Analysis_Retrieval.ipynb    # Modules 1–6
├── app.py                                  # Gradio GUI
├── src/build_corpus.py                     # corpus pipeline
├── src/nlp_pipeline.py                     # Final Pipeline + domain NER (used by the GUI)
├── data/raw/ (pdf, html), data/metadata/source_manifest.csv, data/processed/corpus.json
├── results/    # Tables A–J and checkpoints
│   ├── document_statistics.csv, preprocessing_results.csv (A), tokenization_comparison.csv (B)
│   ├── stem_vs_lemma.csv (C), pos_tagging_results.csv (D), ner_results.csv + ner_entities.csv (E)
│   ├── unigram/bigram/trigram_results.csv, ngram_results.csv (F), bpe_results.csv (G)
│   ├── pipeline_comparison.csv (H), retrieval_results.csv (I), evaluation_results.csv (J)
│   ├── inverted_index.json, queries.csv, relevance_judgments.csv, evaluation_per_query.csv
│   └── pos_gold.csv, ner_gold.csv (annotated gold sets)
├── requirements.txt
└── Report.pdf
```

13. Use of AI assistance

An AI coding assistant (Claude) was used to write and test code, to propose the POS and NER gold labels and the retrieval relevance judgments, and to draft explanatory text. Every proposal was reviewed and approved by the team before use, and all reported numbers come from the notebook outputs and saved CSV files. Design decisions (domain, tokenizer, stop-word policy, pipeline selection) were made by the team on the basis of the experiments reported here.

Team contributions: *(to be completed by the team: who did corpus collection, which modules, the GUI, the report)*

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Full source URLs, retrieval dates and licences for every document: `data/metadata/source_manifest.csv`.